

Lesson 4:



Reminder $y = x\beta + u$

$$y = [x_1; x_2] \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + u$$

$$y = \underset{n \times 1}{x_1} \underset{n \times k_1}{\beta_1} + \underset{n \times k_2}{x_2} \underset{n \times 1}{\beta_2} + u$$

$$b = (x'x)^{-1}x'y$$

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} x_1'x_1 & x_1'x_2 \\ x_2'x_1 & x_2'x_2 \end{pmatrix}^{-1} \begin{pmatrix} x_1'y \\ x_2'y \end{pmatrix}$$

$$P = x(x'x)^{-1}x' \rightarrow Px = x \Rightarrow Pe = 0$$

$$M = I - x(x'x)^{-1}x' \rightarrow Mx = 0$$

$$Me = e \quad MMy = My = My$$

P_1 project on $\ell(x_1)$ M_1 project on $\ell(x_1)^\perp$
 $\ell(x_1) \subset \ell$ $\ell(x_1)^\perp \supset \ell(x)^\perp$

$$P_1 x_1 = x_1 \quad M_1 x_1 = 0$$

$$P_1 x_2 = ? \quad M_1 x_2 = ?$$

$$P x_1 = x_1 \quad M_1 M y = M y = e$$

$$P_2 X_2 = X_2 \quad M_2 X_2 = 0$$

$$P X_2 = X_2 \quad M_2 M_y = M_y$$

$$P_2 X_1? \quad M_2 X_1?$$

$$P_1 = X_1 (X_1' X_1)^{-1} X_1'$$

$$M_1 = I_n - P_1 = I_n - X_1 (X_1' X_1)^{-1} X_1'$$

Frisch Waugh - Lovell (FWL)

$$y = X_1 \beta_1 + X_2 \beta_2 + u$$

$$M_2 y = \frac{M_2 X_1 \beta_1}{y^*} + \frac{M_2 u}{x_1^* + u^*}$$

$$(x_1^{*'} x_1^*)^{-1} x_1^{*'} y^*$$

$$b_1 = (X_1' M_2 X_1)^{-1} X_1' M_2 y$$

$$b_2 = (X_2' M_1 X_2)^{-1} X_2' M_1 y$$

M_y explained P_y residuals

$M_2 y \rightarrow$ regression and $x_2 \rightarrow$ residual

$M_2 X_1 \rightarrow$ regressing X_1 on $X_2 \rightarrow$ residual.

$$b_1 = (X_1' M_2 X_1)^{-1} X_1' M_2 y = (X_1' M_2 X_1)^{-1} M_2 y (X_1 \beta_1 + X_2 \beta_2 + u)$$

$$= \beta_1 + (X_1' M_2 X_1)^{-1} X_1' M_2 y M_2 u$$

$$E(b_1) = \beta_1$$

$$\begin{aligned} \text{Var}_{(b_1)} &= (x_1' M_2 x_1)^{-1} x_1' M_2 u \left[(x_1' M_2 x_1)^{-1} x_1' M_2 u \right]' \\ &= \sigma^2 (x_1' M_2 x_1)^{-1} \end{aligned}$$

$$b_1 \sim N(\beta_1, \sigma^2 (x_1' M_2 x_1)^{-1})$$

$$b_2 \sim N(\beta_2, \sigma^2 (x_2' M_2 x_2)^{-1}) \quad b \sim N(\beta, \sigma^2 (x' x)^{-1})$$

misspecification (1) Omit relevant variables

(2) Include irrelevant var.

$$(1) \quad y = x_1 \beta_1 + x_2 \beta_2 + u \quad (\text{Data gen. process.})$$

$$y = x_1 \beta_1 + u \quad - \text{estimated model}$$

$$\begin{aligned} \hat{\beta}_1 &= (x_1' x_1)^{-1} x_1' y \\ &= \beta_1 + (x_1' x_2)^{-1} x_1' x_2 \beta_2 + (x_1' x_1)^{-1} x_1' u \end{aligned}$$

$$E(\hat{\beta}_1) = \beta_1 + \underbrace{(x_1' x_1)^{-1} x_1' x_2 \beta_2}_{\rightarrow \text{bias}}$$

$$\begin{aligned}\text{Var}(\hat{\beta}) &= E(\hat{\beta} - E(\hat{\beta}))(\hat{\beta} - E(\hat{\beta}))^T \\ &= E((X_1' X)^{-1} X_1' u u' X_1 (X_1' X_1)^{-1}) \\ &= \sigma^2 (X_1' X_1)^{-1}\end{aligned}$$

$$\hat{\theta}_1 \sim N(\beta_1 + (X_1' X)^{-1} X_1' X_2 \beta_2, \sigma^2 (X_1' X_1)^{-1})$$

$$b_1 \sim N(\beta_1, \sigma^2 (X_1' M_2 X_1)^{-1})$$

$$\text{Var}(\hat{\beta}_1) = \sigma^2 (X_1' X_1)^{-1}$$

$$\text{Var}(b_1) = \sigma^2 (X_1' M_2 X_1)^{-1}$$

$$X_1' X_1 \quad X_1' (I - P_2) X_1$$

$$X_1' X_1 \geq X_1' X_1 - X_1' P_2 X_1$$

$$\text{then } (X_1' X_1)^{-1} \leq (X_1' X_1 - X_1' P_2 X_1)^{-1}$$

$$\text{Var}(\hat{\theta}_1) < \text{Var}(b_1)$$

$$\text{MSE}(\hat{\theta}) = E(\hat{\theta} - \theta)^2$$

$$= \text{Var}(\hat{\theta}) + (\text{bias})^2$$

$$E(\hat{\theta} - \theta)(\hat{\theta} - \theta)'$$

$$\text{MSE}(\hat{\beta}_1) = E(\hat{\beta}_1 - \beta)(\hat{\beta}_1 - \beta)'$$

$$= (X_1' X_1)^{-1} X_1' X_2 \beta_2 \beta_2' X_2' X_1 (X_1' X_1)^{-1} + \sigma^2 (X_1' X_1)^{-1}$$

$$MSE(b_1) = \sigma^2 (X_1' M_2 X_1)^{-1}$$

Inclusion of irrelevant variable

$$y = X_1 \beta_1 + u \quad y = X_1 \beta_1 + X_2 \beta_2 + u$$

$$\hat{\beta}_1 = (X_1' M_2 X_1)^{-1} X_1' M_2 y$$

$$\begin{aligned} \hat{\beta}_1 &= (X_1' M_2 X_1)^{-1} X_1' M_2 (X_1 \beta_1 + u) \\ &= \beta_1 + (X_1' M_2 X_1)^{-1} X_1' M_2 u \end{aligned}$$

$$E(\hat{\beta}_1) = \beta_1$$

$$Var(\hat{\beta}_1) = \sigma^2 (X_1' M_2 X_1)^{-1}$$

$$b_1 = (X_1' X_1)^{-1} X_1' y$$

$$E(b_1) = \beta_1 \quad Var(b_1) = \sigma^2 (X_1' X_1)^{-1}$$

$$MSE(\hat{\beta}_1) > MSE(b_1)$$

$$y = X\beta + u$$

$n \times 1 \quad n \times k \quad k \times 1 \quad n \times 1$

Inference hypothesis test

$$\beta_1 = \beta_2, \beta_3 - \beta_5 = 1, \beta_4 + \beta_6 + \beta_7 = 2$$

$= \beta_1 - \beta_2 = 0$

$$H_0: RB = r$$

$$q \times K \quad K \times 1 \quad q \times 1$$

q is # of hypothesis.

$$B^T = [\beta_1 \ \beta_2 \ \dots \ \beta_7 \ \dots \ \beta_K]^T$$

$$R = \begin{pmatrix} 1 & -1 & \dots & \dots & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 & \dots & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_K \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

$R \quad B \quad = \quad C$